The Action-Potential Magnitude Effect:

Typed Waveform, Release, Spectral, Electromechanical, and Cross-Scale Consequences in Self-Aware Networks

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Abstract

Action potentials are all-or-none regenerative events, but successful spikes need not be physically or consequentially identical. Waveform, propagation, terminal calcium, vesicle release, and receiver response can vary with cell type, compartment, channel state, neuromodulation, and recent input history. The unresolved computational problem is how this state-dependent event becomes a receiver-specific consequence without treating waveform, calcium, release, postsynaptic response, and population power as one variable.

I formalize a conditional mediation chain: receptor- and synapse-specific history changes membrane and channel state; that state shapes the cellular event; axons and boutons transform it; calcium and release translate it probabilistically; and a receiver responds according to its immediate state and learned organization. Self-Aware Networks (SAN) calls the computational use of this multidimensional, transformed, receiver-relative event the **Action-Potential Magnitude Effect** (APME). Here *magnitude* names the consequential event vector and its transformations, not peak voltage, duration, calcium, vesicle count, postsynaptic current, or population power alone. SAN further hypothesizes that receiver-relative differences can contribute to phase-wave differentials, Neural Tuning, and distributed neural rendering; those cross-scale roles remain experimental questions.

Version 2 separates electrical waveform, duration, calcium, release, electromechanical deformation, periodic power, and aperiodic spectra as typed measurements; replaces any universal inverse frequency-magnitude claim with preparation-specific local log sensitivities; and reports a deterministic synthetic application plus a bounded Lean 4 kernel. Across 48 systems, seven intervention families, and five models, the correctly typed model achieved intervention-effect RMSE 0.000697 versus 0.038113 for the best alternative. All 15 validators passed, an independent rerun reproduced the declared artifacts, and Lean 4.32.0 compiled eight mathematical leaves with exit code 0 and no admitted proofs. These results establish properties of the declared formal and synthetic models, not biological or cross-scale confirmation. APME earns empirical support only if waveform variables add reproducible, receiver-specific predictive and causal value after relevant covariates are controlled.

Keywords: action-potential waveform; action-potential duration; presynaptic calcium; synaptic release; potassium channels; electromechanical deformation; periodic and aperiodic spectra; local log sensitivity; phase-wave differential; neural tuning; neural rendering; Self-Aware Networks

1 Introduction

1.1 The ordinary problem

Consider two successful spikes from the same neuron. Both cross threshold. One follows a different conductance history, reaches a bouton with a different channel state, and acts on a receiver in a different state. Calling both events *spikes* is correct, but it does not answer whether their downstream consequences are equivalent. The ordinary problem is therefore not whether a spike occurred. It is how a stateful source, a stateful transmission path, and a stateful receiver turn that occurrence into a particular consequence.

1.2 The established scientific account

The all-or-none principle states that once a regenerative action potential is successfully initiated, it is not a graded copy of the initiating stimulus. A stronger claim is often inferred: that all successful spikes are equivalent symbols whose only relevant variables are occurrence and timing. That inference is not required by the physiology.

In cortical pyramidal neurons driven by naturalistic conductance patterns, somatic spike height and width have carried information about tens of milliseconds of preceding input history [1]. At presynaptic terminals, potassium and sodium channel state can change spike amplitude or duration, and those changes can alter calcium entry and synaptic transmission [2-7,10-13]. The result is neither a purely analog neuron nor a rejection of regenerative spiking. It is a hybrid system: discrete spike initiation embedded in continuously changing cellular and synaptic state.

1.3 The remaining computational problem

Established physiology supplies many components, but naming those components does not yet specify a computational operator. A usable account must keep the variables typed, show where information can be transformed or lost, identify parallel routes, and state what a particular receiver can distinguish. It must also explain why waveform-aware prediction is not merely a relabeling of rate, timing, phase, or stimulus history.

1.4 The operation in plain language

The proposed operation is a sequence of physical transformations linking a recognized or unexpected input to a receiver-relative consequence:

```
receptor- and synapse-specific input history
-> membrane and ion-channel state
-> somatic, axonal, and terminal waveform
-> terminal calcium entry
-> probabilistic vesicle release
-> postsynaptic response
-> receiver-relative phase-wave differential
-> changed population routing, rendering, or learning
```

Each arrow is conditional. Each stage has distinct units. A duration difference need not survive propagation; a terminal waveform difference need not alter release; altered release need not change the receiver; and a receiver change need not affect perception or behavior. These possible failures are not peripheral qualifications. They define the scientific content of APME.

1.5 The vocabulary is earned

SAN names the computational use of this richer, transformed event the **Action-Potential Magnitude Effect**. The term is introduced only after the problem, inherited physiology, and proposed operation are explicit. *Magnitude* is intentionally multidimensional and receiver-relative. It does not rename phase coding, rate coding, traveling waves, waveform modulation, or synaptic release. Those established descriptions can supply components or alternatives; APME asks whether their joined source-to-receiver transformation adds a distinct, testable explanatory variable.

2 What APME Claims

2.1 Minimal claim

The minimal APME claim is:

> A successful action potential can retain measurable effects of the source neuron's recent state in its waveform, and a named downstream receiver may be affected by a conditionally transformed consequence of that waveform.

This claim is compatible with established physiology. It does not by itself establish SAN, PWD, NAPOT, qualia, or consciousness.

2.2 SAN mechanistic hypothesis

The stronger SAN hypothesis is:

> Sensory recognition or novelty can alter receptor, synaptic, and membrane state in ways that produce multidimensional changes in spike timing, duration, waveform, bursts, terminal calcium, and synaptic release. When interpreted relative to a receiver's expected or tonic state, those changes can function as PWD events that modify effective routing and distributed neural rendering.

The words *recognition* and *novelty* refer to experimentally declared contrasts, not intrinsic labels contained inside a spike. The proposal does not predict that every recognized or novel stimulus broadens every action potential. Depending on cell, compartment, state, and pathway, a stimulus may widen, narrow, delay, advance, suppress, burst, or leave the waveform unchanged.

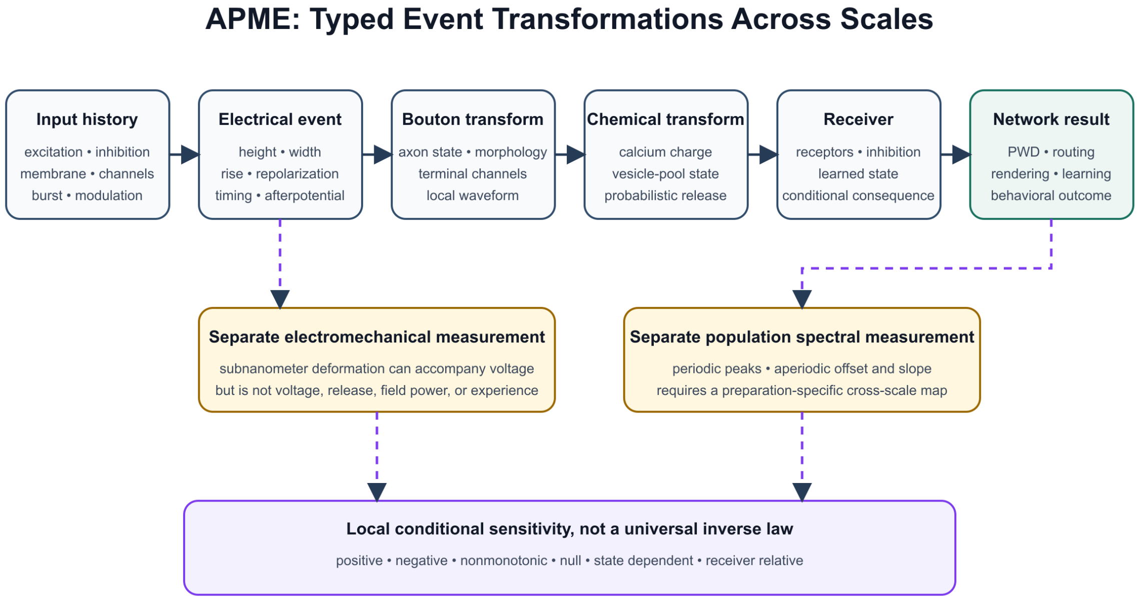

Figure 1. The APME chain keeps electrical waveform, chemical release, electromechanical deformation, receiver response, and population spectra as distinct measurement types. Dashed arrows require empirical cross-scale maps.

2.3 Operational distinctions

APME is distinguished by a typed transformation chain rather than by treating any single measurement as the code. Peak voltage is one waveform feature, not a universal stimulus-magnitude code. Somatic width is transformed during propagation and therefore is not copied unchanged into synaptic-current duration or a deterministic vesicle count. Release alphabets are synapse-specific and probabilistic. Population `1/f` spectra and single-neuron waveforms require different measurements. Percepts and memories are distributed consequences rather than contents of one spike, and the participating network performs the observer operation rather than reporting to one observing neuron. The decisive proposition is consequently conditional: a longer spike may matter in a named pathway and state, but recognition, novelty, value, or consciousness cannot be read universally from duration alone.

3 Measurement Types and Scale Boundaries

3.1 Action-potential waveform

For a recorded spike `i` in compartment `c`, define a typed waveform feature vector

$$\mathbf{w}_{c,i} = \left[V_{\text{peak}}, A_{\text{AP}}, \tau_{1/2}, s_{\uparrow}, s_{\downarrow}, AHP, ADP, t_i\right]_{c,i}. \tag{1}$$

Here `V_peak` is peak voltage, `A_AP` is amplitude relative to a declared baseline or threshold, `tau_1/2` is half-width, `s_up` and `s_down` are rise and repolarization measures, `AHP` and `ADP` describe afterhyperpolarization and afterdepolarization, and `t_i` is event time. The compartment and recording method must be stated because somatic and terminal waveforms need not be identical [5,6].

3.2 Conductance and state history

Let the pre-spike history available to source neuron `s` over window `W_H` be

$$\mathbf{h}_{s,i}(W_H) = [g_E(t), g_I(t), V_m(t), \mathbf{o}_{Na}(t), \mathbf{o}_K(t), \mathbf{m}(t), T, \mathbf{b}(t)]_{t\in[t_i-W_H,t_i)} . \tag{2}$$

The vector includes excitatory and inhibitory conductances, membrane voltage, sodium- and potassium-channel availability, neuromodulatory state, temperature, and recent burst history. It is not expected that every experiment can measure every term. The unmeasured terms become explicit latent variables rather than being silently assigned to duration.

The physiological history-to-waveform map is

$$\mathbf{w}_{c,i} = F_c(\mathbf{h}_{s,i}, \boldsymbol{\theta}_s) + \boldsymbol{\epsilon}_{c,i}, \tag{3}$$

where `theta_s` includes cellular morphology and relatively stable channel expression. De Polavieja and colleagues provide an important component result: in their preparation, spike waveform carried reliable information about preceding conductance history [1]. Equation (3) is a system-identification statement, not a claim that the complete input is recoverable from one spike.

3.3 Population `1/f` is a different measurement

For EEG, MEG, ECoG, or local field potential, an aperiodic power spectrum is often approximated by

$$P(f) = Af^{-\beta} + P_{\text{osc}}(f) + \eta(f). \tag{4}$$

This equation describes power spectral density across frequency. `A` and `beta` characterize a population-level field measurement under a declared preprocessing model. Miller and colleagues, for example, measured power-law scaling in human ECoG [8]. Equation (4) is not an inverse law between the peak amplitude and duration of individual action potentials. APME treats possible cellular-to- population relations as hypotheses requiring simultaneous multiscale recording, not as an identity.

4 Conditional Synaptic Transformation

4.1 Terminal waveform and calcium

Let a somatic or AIS spike propagate to bouton `b` through transformation `G_b`:

$$\mathbf{w}_{b,i} = G_b(\mathbf{w}_{AIS,i}, \mathbf{a}_{b,i}, \boldsymbol{\theta}_b) + \boldsymbol{\xi}_{b,i}, \tag{5}$$

where `a_b,i` represents axonal and bouton state. Potassium and sodium channels can shape the presynaptic waveform, and their effects vary across synapses and firing regimes [2-7]. Therefore a somatic duration measurement is not a substitute for a bouton recording.

Calcium entry is then

$$Q_{Ca,b,i} = \left| \int_{t_i^-}^{t_i^+} I_{Ca,b}\big(V_b(t), \mathbf{o}_{Ca,b}(t), [Ca^{2+}]_o, T\big)\, dt \right| . \tag{6}$$

The absolute value declares calcium charge magnitude under the common convention that inward current is negative; studies using another convention must state it. The falling phase can be especially consequential because channel activation and calcium driving force change during repolarization [3,4]. Wider spikes can increase calcium and release in several preparations [2,3,5,6,11,13], but waveform-stabilization and cancellation mechanisms also exist [7,12]. Release can also change through release-machinery state without a measured change in presynaptic waveform or calcium influx [14]. Thus APME predicts a family of conditional transfer functions, not a monotonic or exclusive pathway.

4.2 Probabilistic vesicle release

For `N_RRP` release-ready vesicles, a simple bounded release model is

$$K_{b,i} \mid N_{RRP,b,i}, p_{rel,b,i} \sim \text{Binomial}(N_{RRP,b,i}, p_{rel,b,i})\,, \qquad p_{rel,b,i} = H_b(Q_{Ca,b,i}, d_b, \mathbf{z}_{b,i})\,, \tag{7}$$

where `K_b,i` is released vesicle count, `d_b` describes channel-sensor geometry, and `z_b,i` includes residual calcium, depletion, facilitation, inhibition, and neuromodulation. Equation (7) is a minimal baseline whose independent, exchangeable release-site assumption must be tested. Correlated or heterogeneous sites require a beta-binomial, Poisson-binomial, or mechanistic release-site model. The equation replaces a fixed zero-to-three-vesicle alphabet with a synapse-specific probabilistic family. The historical SAN wording remains important genealogy, but the scientific model must allow different readily releasable pools, multivesicular release, failures, reluctant vesicles, and preparation-specific coupling [14].

4.3 Receiver response

The postsynaptic consequence is

$$\mathbf{y}_{r,i} = R_r\big(K_{b,i}, q_{b,i}, \mathbf{x}_r(t_i^-), \boldsymbol{\theta}_r\big) + \boldsymbol{\nu}_{r,i}. \tag{8}$$

Here `q_b,i` is quantal size, transmitter identity, or another declared per-release input rather than a second vesicle-count term; `x_r` is the receiver's immediate state, and `theta_r` is its anatomical and learned response function. APME is receiver-relative: the same presynaptic event can have different consequences at different boutons and receivers.

5 From APME to Phase-Wave Differentials

APME supplies one candidate physical source of PWD events. It is not coextensive with PWD. A PWD may arise from excitation, inhibition, timing, phase, frequency, duration, transmitted quantity, or another bounded departure from a declared reference state.

For receiver `r`, define a tonic or expected reference

$$\mathbf{b}_r(t; W_R) = \mathbb{E}[\mathbf{x}_r(t) \mid \mathcal{C}, W_R]\,, \tag{9}$$

where `C` names task, behavior, preparation, reference signal, and state. The receiver-relative event is

$$\begin{aligned}\mathbf{d}_{r,i} &= [d_{\text{circ}}(\phi_i, \phi_{0,r}), \Delta f, \Delta A, \Delta\tau, \Delta q, \Delta CV, \Delta\mathbf{x}_r]_i\,,\\ \mathbf{p}_{r,i} &= \mathbf{D}_r^{-1}\mathbf{d}_{r,i}.\end{aligned} \tag{10}$$

Here `d_circ` is a declared circular phase distance and `D_r` is a preregistered diagonal matrix of nonzero coordinate scales, such as baseline standard deviations or measurement tolerances. The raw typed vector `d_r,i` preserves units; the normalized vector `p_r,i` permits comparison without silently adding milliseconds, volts, radians, and vesicle-related quantities. Normalization does not prove that the coordinates form one biological code. Duration is one typed coordinate, not the token as a whole. A measured vector becomes a candidate biological token only if a receiver preserves or systematically transforms the distinction, the vector adds held-out predictive value, and selective perturbation changes the predicted consequence.

For positive inter-event intervals, SAN's later coefficient-of-variation extension can be written

$$CV_{ISI} = \frac{\sigma_{ISI}}{\mu_{ISI}}. \tag{11}$$

Ordinary CV is not suitable for circular phase and does not uniquely characterize waveform variability. APME experiments should compare CV with local variation, CV2, Fano factor, circular variance, waveform covariance, and conditional history models.

6 Recognition, Novelty, and Qualia as Test Domains

6.1 Whisker touch

Whisker behavior provides a tractable sensorimotor test. Active whisking, passive touch, active contact, object location, expectation, and novelty can be separated while membrane potential, spikes, movement, and population state are measured. Existing barrel-cortex work shows strong state-dependent sensory processing [9]. It does not establish a universal AP-duration signature.

An APME experiment should ask whether matched contact events produce reproducible waveform or terminal differences after controlling whisker kinematics, cortical state, cell identity, spike count, rate, and timing. The prediction is not simply `novel > familiar` duration. The prediction is that a multidimensional waveform-aware model may add information about history and receiver effect.

6.2 Umami

Umami provides a receptor-specific pathway in which transduction, adaptation, internal state, and learned recognition can be manipulated. SAN's 2022 umami note proposed a route from taste receptor signaling through potassium and calcium regulation to output duration and release. This is a mechanistic hypothesis. Taste transduction is distributed across receptor cells, afferents, brainstem, thalamic, insular, orbitofrontal, homeostatic, and motor pathways; it is not one generic neuron receiving the semantic label *umami*.

The decisive experiment would compare expected, unexpected, learned, and unlearned taste stimuli while measuring identified pathway stages. A characteristic signature must generalize across held- out trials and predict downstream effects better than intensity, movement, licking, arousal, firing rate, and spike timing alone.

6.3 Color and multimodal objects

SAN uses redness and broccoli to explain distributed representation. Redness is not stored in one duration value, and broccoli is not reconstructed from one spike. Color, shape, taste, texture, smell, body relation, value, and action affordance are represented across pathways and populations. APME proposes that waveform-sensitive cellular events may contribute to the changing differences within that distributed state.

In SAN terminology, the ongoing network state is the tonic canvas and differentiated events are phasic ink. These are proposed operational roles, not decorative metaphors and not a literal screen. The participating recurrent network is the distributed observer-action process. No localized inner viewer receives the result.

7 APME, Neural Tuning, and the Gamma Consideration Sandwich

Neural Tuning describes changes in a receiver's response readiness or effective routing. Let

$$\begin{aligned} \mathbf{x}_{r,i}^{+} &= S_r\left(\mathbf{x}_{r,i}^{-}, \mathbf{p}_{r,i}, \mathbf{c}_i\right), \\ \boldsymbol{\theta}_{r,n+1} &= U_r(\boldsymbol{\theta}_{r,n}, \mathbf{p}_{r,i}, \mathbf{c}_n, \mathbf{y}_{r,i}) . \end{aligned} \tag{12}$$

The first line is a transient receiver-state update; the second is a slower parameter or plasticity update. Temporary effective connectivity can depend on either, but an immediate change in `x_r` must not be mislabeled as persistent learning in `theta_r`. These outcomes must be distinguished experimentally.

Within the Gamma Consideration Sandwich, Layer-4 alpha carries primary-sensory-side input and Layer-4 beta carries prefrontal thought/top-down input in the top bun. Layer-2/3 gamma performs the proposed consideration and proprioceptive-coordination operation in the functional middle. Layer-5 theta/gamma carries action-side output and returning-body input in the bottom bun, with body evidence reaching the cortical circuit through established ascending pathways. APME supplies a possible cellular transformation inside that architecture: sensory and thought history shape cellular state; cellular state shapes transmitted events; and receiver consequences alter the next recurrent cycle. The GCS mapping is a target-specific hypothesis, not a claim that one cortical layer or frequency performs the same computation everywhere.

8 Historical Genealogy and Priority Boundary

The current source record supports a staged genealogy:

Stage	Source	Bounded contribution
2017	Neural Lace/SAN architectural material	Dendritic computation, coincidence, distributed representations, alternating transmission, and three- or four-dimensional activity patterns. These are ancestors, not the mature APME operator.
June 8, 2022	`a0089z ctpr.txt`	Public Git fixation of the all-or-none/variable-duration question and an explicitly speculative potassium -> AP duration -> terminal calcium-open time -> release chain. This is an early joined hypothesis, not yet the APME name or a validated law.
July 7-17, 2022	`a0149z.md`, `a0115z.md`	Neuronal duration/frequency patterns proposed as contributors to population signals, followed by the explicit magnitude/frequency cross-scale question.
August 3-10, 2022	`a0053z.md`, `a0008z.md`, `a0272z.md`	The name Action Potential Magnitude Effect; sharper potassium/waveform/calcium/release wording; and an umami/cAMP application.
August-September 2022	rendering and NAPOT revisions, including `a0007z.md`	APME connected to phase variation, array projection, and neural rendering; bounded vesicle-output language introduced.
November 2022	`a0296zWhisker.md`	Whisker input connected mainly to state, desynchronization, coincidence, and rendered tactile differentiation. It does not establish universal broadening.
January 2024	`a0011z.md`	Explicit mechanoreceptor-to-potassium-to-duration hypothesis.
September 2024	`02san.md`	PWD operator developed as a multidimensional departure from tonic state and connected to Neural Tuning and rendering.
June 2025	`13san.md`	Duration and recent variability developed as coefficient-of-variation-like writable symbols.

Targeted Git history distinguishes two priority statements. The joined speculative chain is already present in the June 8, 2022 `a0089z ctpr.txt` blob (`37da575ba33f85cef4239295be161066d8d39f4c`, lines 19-33). The coined term **Action Potential Magnitude Effect** first appears in the August 3 commit (`61d4fe1d03b158d0329361976b0178d0e5d43350`, line 5). The sharper sentence asserting that potassium state determines wave shape and calcium-channel open duration appears on August 5 (`a214ccf6d9ceb117638ce5fb2cb5cbabe6f73692`, lines 23-25). The source record therefore supports June 8 for the joined hypothesis, August 3 for the name, and August 5 for the mature wording; none of those dates turns the mechanism into an established physiological result.

The priority boundary is equally important. Earlier science established all-or-none spiking, variable waveforms, potassium-channel repolarization, calcium-triggered exocytosis, probabilistic release, analog-digital modulation, phase coding, and population power laws. SAN's documented delta is the joined, receiver-relative architecture linking sensory history, channel state, waveform, release, PWD, tuning, and distributed rendering. The distinction is functional: the inherited components describe measured phenomena, while APME proposes how their conditional transformations become a typed computational event for a particular receiver.

8.1 Retroactive intent recovery

The June and July 2022 records contain a cross-scale question about an inverse relationship between frequency and *magnitude*. That wording must be preserved as historical evidence, but it should not be silently promoted into a universal law. EEG/ECoG `1/f` describes population spectral power across frequency; single-neuron spike width and amplitude are different observables. The recovered legitimate problem behind the early wording is whether limited, stateful cellular and synaptic resources create systematic transformations among event rate, waveform, and downstream effect.

Version 2 therefore corrects the measurement identity while preserving the research direction. It turns the early question into simultaneous multiscale experiments and allows positive, negative, nonmonotonic, stabilizing, or null relations. This is an intent recovery, not a backdated claim: the original 2022 wording remains fixed to its original source, and the typed correction is dated to this manuscript.

9 Experimental Program

9.1 Experiment 1: waveform history replication

Repeat and extend naturalistic dynamic-conductance experiments in identified adult neurons at physiological temperature. Compare models using:

1. spike count and instantaneous rate;
2. spike timing;
3. waveform features;
4. history-conditioned waveform features; and
5. full subthreshold voltage.

Use held-out stimuli, cells, and animals. APME requires reproducible incremental information from waveform variables, not merely within-sample decoding.

9.2 Experiment 2: source-to-bouton mediation

Simultaneously measure soma or AIS waveform, bouton voltage, bouton calcium, release, and postsynaptic response. Manipulate potassium or sodium channel state while holding spike count and timing as constant as feasible. Estimate the mediated effects

$$\mathcal{M}_{APME} = \{\mathbf{w}_{AIS} \to \mathbf{w}_{bouton} \to Q_{Ca} \to K \to \mathbf{y}_r\}\,. \tag{13}$$

The chain is supported only when measured mediators explain the later effect and intervention on a mediator changes the predicted downstream consequence.

9.3 Experiment 3: recognition and novelty

Use whisker, taste, visual, or auditory tasks with expected, repeated, deviant, learned, and novel conditions. Match physical stimulus energy and control movement, arousal, attention, reward, adaptation, firing rate, spike timing, cell type, and baseline membrane state. Pre-register whether the candidate signature is wider, narrower, multivariate, or pathway-specific.

9.4 Experiment 4: receiver specificity

Record multiple boutons from one axon and multiple postsynaptic targets. Test whether the same source event has different effects according to bouton and receiver state. Receiver specificity is a central APME prediction and distinguishes the proposal from a scale-free conserved magnitude.

9.5 Experiment 5: multiscale bridge

Record cellular waveforms and population fields simultaneously. Test whether cellular waveform features predict later population-state changes after spike timing, rate, synaptic currents, and behavioral state are included. Do not infer the bridge from shared words such as *amplitude*, *duration*, or *frequency*.

10 Model Comparison and Statistical Tests

Let `Y` be a declared receiver, population, perceptual, or behavioral outcome. Compare nested models:

$$\begin{aligned}
&M_0 : Y \sim \text{stimulus} + \text{state},\\
&M_1 : M_0 + \text{spike count/rate},\\
&M_2 : M_1 + \text{spike timing/phase},\\
&M_3 : M_2 + \mathbf{w}_{AIS},\\
&M_4 : M_3 + \mathbf{w}_{bouton} + Q_{Ca} + K,\\
&M_5 : M_4 + \mathbf{x}_r + \text{interactions}.
\end{aligned} \tag{14}$$

The relevant result is out-of-sample improvement, calibrated uncertainty, and causal mediation, not the significance of one duration coefficient. Nonlinear models should be compared with interpretable baselines.

Analyses must avoid leakage from repeated trials, cell identity, post-outcome variables, or preprocessing choices.

An APME effect size can be operationalized as

$$\Delta I_{APME} = I(Y; \mathbf{w}, Q_{Ca}, K \mid \text{stimulus,state,count,timing}) \,, \tag{15}$$

or as an equivalent cross-validated predictive improvement. Positive conditional information alone does not prove mechanism; perturbation and rescue are required for the causal chain.

10.1 Competing routes and identifiability

The proposed waveform route sits inside a larger causal system. Release machinery, readily releasable-pool state, channel-sensor geometry, quantal size, receptor state, inhibition, and neuromodulation can alter the same outcome. A valid test therefore begins with the expanded graph:

```
history -> soma/AIS waveform -> bouton waveform
-> calcium -> release -> receiver outcome
history ---------------------> release machinery -------------> receiver outcome
vesicle-pool and geometry -----------------------> release ----> receiver outcome
receiver state, inhibition, and receptor state ----------------> receiver outcome
```

Let `W` be the manipulated waveform feature vector and let `Y` be the declared downstream outcome. The total interventional contrast is

$$\Delta_Y^W = \mathbb{E}[Y \mid do(\mathbf{W} = \mathbf{w}_1)] - \mathbb{E}[Y \mid do(\mathbf{W} = \mathbf{w}_0)] \,. \tag{16}$$

This contrast is not attributable to the APME route until the measured mediators and parallel routes are identified. Observational associations, including strong decoding, remain insufficient when an unmeasured common state could cause both waveform and outcome.

10.2 Incremental predictive criterion

Let `M_alt` include stimulus, behavior, membrane state, count, rate, timing, phase, release-state, pool, receptor, and receiver covariates available in the preparation. Let `M_full` add the source-to-bouton waveform, calcium, release, and receiver interactions specified by APME. The pre-registered held-out criterion is

$$\Delta S_{\text{APME}} = S_{\text{CV}}(M_{\text{full}}) - S_{\text{CV}}(M_{\text{alt}}) > 0. \tag{17}$$

The score, minimum effect, uncertainty interval, and generalization unit must be declared in advance. Generalization across trials alone is inadequate if cell, animal, session, or stimulus identity leaks between training and test folds.

10.3 Perturbation, blockade, and rescue

For a nominated mediator `M`, mechanistic support requires more than a positive total contrast. The perturbation should change the downstream outcome; selectively blocking the mediator should attenuate the predicted effect; and restoring the mediator while the initiating manipulation is held should restore the predicted consequence. A bounded acceptance set is

$$\mathcal{C}_{\text{rescue}} = \left\{ |\Delta_Y^W| > \delta_Y, \quad |\Delta_Y^{W|block(M)}| \leq \delta_0, \quad |\Delta_Y^{rescue(M)} - \Delta_Y^W| \leq \delta_R \right\}. \tag{18}$$

The tolerances are preparation-specific and must be pre-registered. Failure at a particular arrow localizes where the proposed information is lost or bypassed instead of being redescribed as a global success or failure.

10.4 Typed event and spectral quantities

The first public version separated population `1/f` power from the waveform of one action potential. Version 2 strengthens that separation into a typed measurement contract. Let the event descriptor for spike `i` at compartment `c` be

$$\mathbf{e}_{c,i} = \left(\mathbf{w}_{c,i}, Q_{Ca,c,i}, K_{c,i}, \boldsymbol{\delta}_{c,i}^{mech} \right), \tag{19}$$

where the waveform vector `w` contains electrical quantities, `Q_Ca` is calcium charge, `K` is a release variable, and `delta_mech` is a separately measured deformation field or summary. These coordinates do not share units and cannot be added without a declared normalization and question. Mechanical motion accompanying action potentials has been optically measured at subnanometer scale in mammalian neurons [23,24]. Those measurements support an additional observable; they do not show that deformation substitutes for voltage, release, field power, or experience.

For a population recording, define a typed spectral representation

$$P(f,t) = P_{aper}(f,t) + \sum_{k=1}^{K} P_k^{osc}(f,t) + \eta(f,t), \tag{20}$$

with periodic peaks separated from the aperiodic component under a declared parameterization. Donoghue and colleagues formalized this practical separation [15]. Current biophysical modeling identifies multiple possible contributors to the aperiodic background and cautions against a single physiological interpretation of its exponent [16]. Pharmacological and interneuron manipulations also fail to support one universally reliable exponent-to-excitation/inhibition mapping [17]. APME therefore treats slope, offset, oscillatory peak parameters, and event-level variables as typed measurements whose cross-scale relation must be learned in a specific preparation.

10.5 Local cross-scale sensitivity

The early SAN sources asked whether frequency and magnitude might bear an inverse relation. The source-faithful scientific recovery is not a universal power law. It is a local, conditional sensitivity question. For a positive response `y` and positive intervention variable `x`, define

$$S_{y\leftarrow x}(\mathbf{z}) = \frac{\partial \log y}{\partial \log x} = \frac{x}{y}\frac{\partial y}{\partial x}, \tag{21}$$

at state `z`. This dimensionless elasticity is invariant to nonzero changes of measurement units. It can be positive, negative, nonmonotonic across states, or locally zero. It therefore tests the recovered rate-magnitude question without assuming the answer.

For a typed path `x -> m -> y`, the local chain rule is

$$S_{y\leftarrow x} = S_{y\leftarrow m} S_{m\leftarrow x}, \tag{22}$$

when the required derivatives and nonzero denominators exist and the nominated mediator captures the path being analyzed. With parallel routes, the total sensitivity is not generally the product of one chain. It must be decomposed or estimated under interventions.

This framework accommodates known biological diversity. Presynaptic waveform plasticity can alter calcium entry and release [5]. Kv-beta-1 can control facilitation [18]. Patient-derived LGI1 autoantibodies have been reported to reduce presynaptic Kv1 surface density, broaden presynaptic action potentials, and enhance transmission in the studied preparations [19]. Subthreshold electric fields can produce polarity- and morphology-dependent changes in axonal voltage, calcium, and glutamate release [20]. Distinct active-zone machineries contribute differently to calcium- channel clustering and vesicle priming [21], while reconstituted minimal machinery separates fast and delayed release sensors, clamp behavior, residual calcium, and activity history [22]. These results support a typed, state-dependent pathway; they rule against collapsing all effects into one universal scalar.

10.6 Cross-scale comparison and the QGTCD boundary

Let `C` be a declared map from event-level features to a population observable. A cross-scale claim has the form

$$\boldsymbol{\pi}(t) = C_{\mathcal{P},\mathcal{M}}\left(\{\mathbf{e}_{c,i}\}_{i,c}, \mathbf{z}(t)\right) + \boldsymbol{\epsilon}(t), \tag{23}$$

where the preparation `P`, measurement pipeline `M`, state `z`, and error model are explicit. This form allows simultaneous intracellular, presynaptic, optical, and population recordings to test which event features predict which population parameters. It forbids an identity arrow from spike duration to aperiodic exponent unless the map is independently established.

SAN's 2022 notes later intersected with questions developed under Quantum Gradient Time Crystal Dynamics (QGTCD) about magnitude-frequency relations across scales. That genealogy is preserved as a separately bounded research bridge. Neither the present physiology, the synthetic application, nor the Lean kernel derives gravitational dynamics from neural data, validates QGTCD, or shows one universal exponent across neural and gravitational systems. Any future comparison must first type the quantities and units in each domain, identify a dimensionless invariant, state the projection map, and test it independently. Formal resemblance is not physical identity.

10.7 Bounded formal verification

The version-2 Lean kernel machine-checks eight leaves:

1. frequency times its declared nonzero period equals one;
2. period does not determine amplitude;
3. an algebraic local log-sensitivity expression is invariant under unit rescaling;
4. local sensitivities compose along a two-link chain under the declared assumptions;
5. local exponents need not be universal;
6. the same source can produce different consequences at receivers with different gain;
7. restoring a declared mediator restores the modeled outcome; and
8. a finite counterexample refutes a universal inverse-rate law.

Lean 4.32.0 and Mathlib v4.32.0 compiled the current source with exit code 0. A whole-word scan found no `sorry` or `admit`. Seven real-algebra leaves use Lean's standard quotient, choice, and propositional-extensionality foundations; the substrate-agnostic mediator-rescue leaf reports no axioms. The exact command, source hash, and theorem-level output are preserved in `formal/FORMAL-CHECK-RECEIPT-20260825.md`.

These theorems protect measurement logic. They do not prove that any preparation instantiates the model, that an observed association is causal, that a universal neural law exists, or that a neural-to-gravitational map has been found.

11 Deterministic Reference Application

11.1 Frozen design

The application generates 48 synthetic systems balanced across four true rate-magnitude relation families: positive, negative, nonmonotonic, and null. Each system contains a typed route from rate, waveform state, and receiver state through a mediator to a downstream endpoint and an associated synthetic population-spectrum measurement. Five models are compared across seven held-out intervention families, producing 1,680 model-intervention rows:

1. rate-only;
2. waveform-only;
3. untyped scalar "magnitude";
4. a larger endpoint black box; and
5. a typed mediation model.

The test suite includes relation changes, mediator interventions, receiver-state changes, cross-scale perturbations, correct rescue, and controls against a forced inverse law. Configuration, seeds, generator, models, metrics, and validators were frozen before the reported run.

11.2 Results

Mean intervention-effect RMSE was:

Model	Mean effect RMSE
Rate-only	0.167779
Waveform-only	0.169231
Untyped scalar magnitude	0.167676
Endpoint black box	0.038113
Typed mediation model	**0.000697**

The typed-to-best-alternative error ratio was 0.01828. The typed model classified the sign/family of the local relation with accuracy 1.0, recovered the sign of local sensitivity with accuracy 1.0, and achieved sensitivity RMSE 0.000312. Its spectral-exponent RMSE was 0.000160 within the declared synthetic cross-scale map.

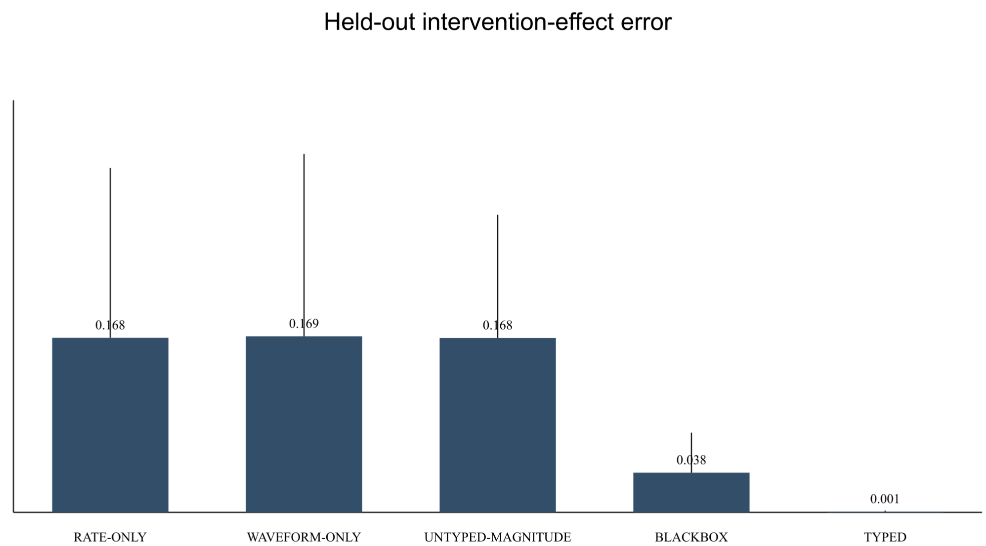

Figure 2. Intervention-effect RMSE for the five models. The typed model receives variables defined by the generator; this is a test of identifiability under the declared measurements, not a claim that biological systems expose those variables without error. Bars are descriptive synthetic summaries and make no tissue, animal, or population inference.

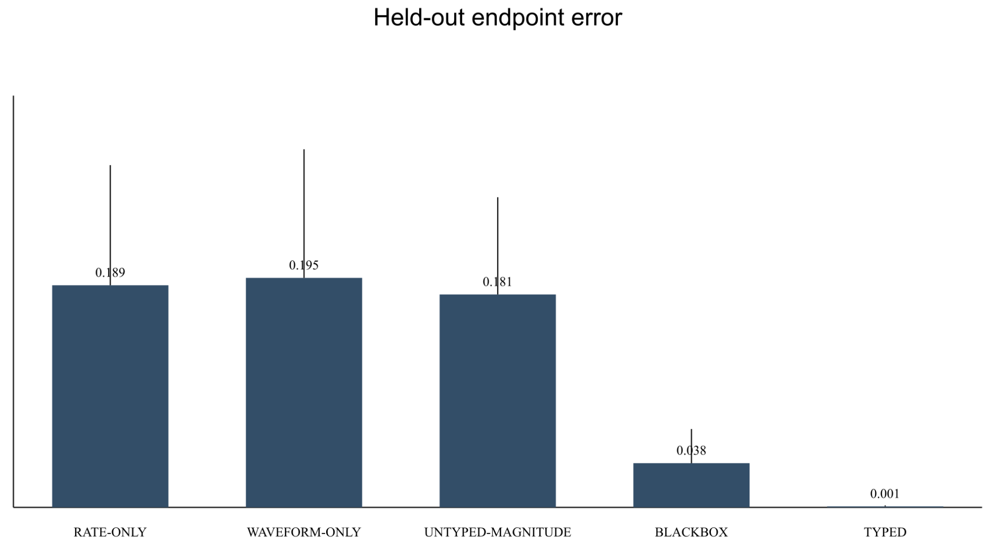

Figure 3. Endpoint RMSE across the frozen held-out interventions. Means and standard deviations describe the frozen synthetic systems only.

A model forced to impose the historical inverse-rate relation was decisively worse when the true synthetic family was positive, nonmonotonic, or null. Its error divided by the correctly typed model's error had a minimum of 37.26 and a median of 303.98. This result does not empirically refute an inverse relation in any biological preparation. It shows why the paper must estimate the local relation rather than encode a universal exponent.

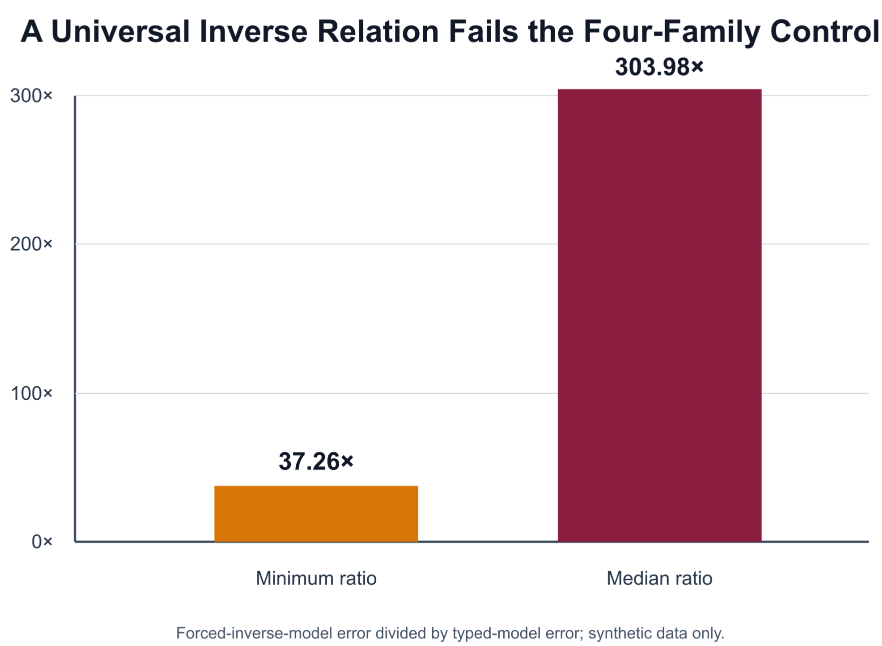

Figure 4. Error ratio for the forced inverse relation relative to the correctly typed model. The control is synthetic and does not test any living preparation.

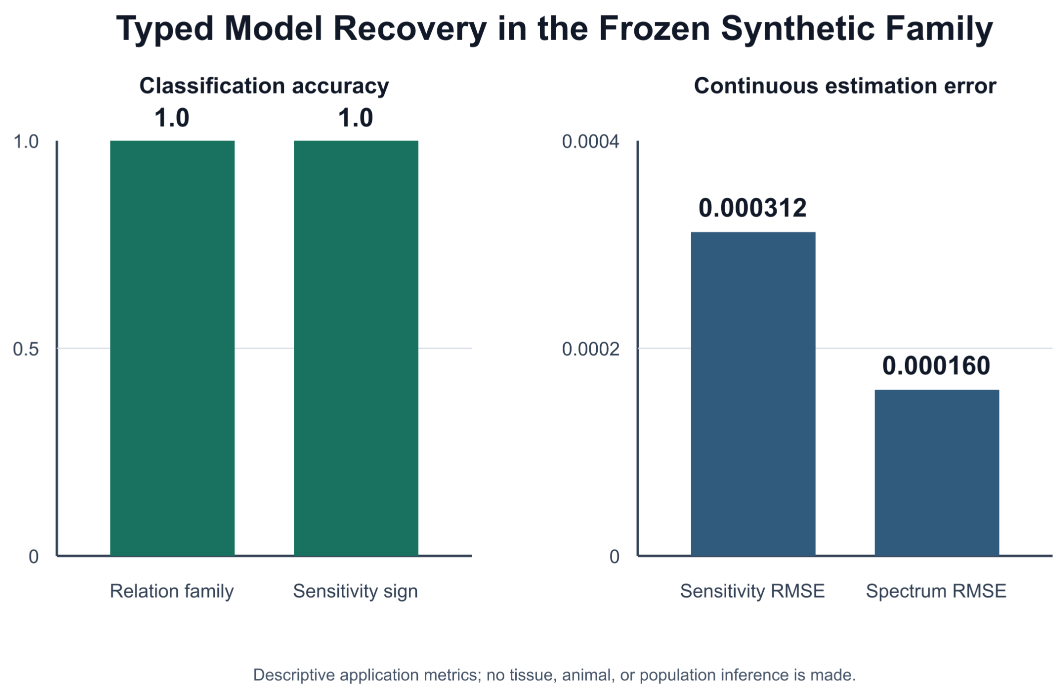

Figure 5. Recovery of relation family, local sensitivity sign, sensitivity value, and synthetic spectral exponent. Perfect classification and small RMSE arise in a known, fully observed generator; they are not biological estimates.

11.3 Validation and determinism

All 15 validators passed. They check the frozen configuration, code and artifact hashes, row counts, model and intervention completeness, held-out performance, relation and sensitivity recovery, inverse-law control, cross-scale output, finite values, provenance, and determinism. A second run reproduced every declared artifact byte-for-byte.

The result is an executable proof of concept for typed mediation and local sensitivity. The generator's ground truth is known, hidden biological confounding is absent, and the typed model is well matched to the data-generating process. A future application must add measurement noise, hidden state, stochastic release, misspecified mediators, varying observability, partial interventional training, and stronger system-identification baselines.

11.4 What the application supports

Within this frozen family, one untyped magnitude variable cannot represent positive, negative, nonmonotonic, and null routes at once. A correctly typed mediation model can recover the local relation and predict held-out interventions. A forced inverse law fails outside the family it assumes. A declared cross-scale map can be estimated and audited when its measurements are available.

The application does not establish that APME is biologically sufficient, that all relevant mediators have been measured, that the cross-scale map exists in vivo, or that the reported error ratios will generalize to neural data.

12 Falsifiers

The joined APME hypothesis is weakened or rejected in a declared preparation if:

1. waveform features are not reproducible after measurement noise and state are controlled;
2. waveform features add no held-out information beyond count, rate, timing, and subthreshold state;
3. somatic differences do not reach the relevant boutons;
4. bouton waveform differences do not alter calcium or release;
5. measured calcium and release fully explain the effect and no receiver-relative APME variable adds value;
6. recognition or novelty effects disappear after stimulus, movement, adaptation, and arousal controls;
7. the same event has no receiver-specific transformation where the theory predicts one;
8. perturbing the proposed channel or waveform mediator does not change the predicted consequence;
9. rescue of the mediator does not restore the effect; or
10. population or behavioral effects are explained equally well by simpler timing- or rate-only models.

A null result at one synapse does not disprove waveform-dependent transmission everywhere. It does disprove an unqualified universal APME claim. The theory should therefore state the cell, pathway, state, receiver, and timescale before testing.

13 Discussion

APME reframes a familiar dichotomy. Neural signaling is neither adequately described as purely binary nor safely summarized as continuously graded. Spike initiation is regenerative, but the event occurs in a stateful cell, propagates through a stateful axon, enters stateful boutons, and is interpreted by stateful receivers. Biology can therefore combine discrete event generation with analog and probabilistic transformation.

The proposal's scientific value depends on disciplined typing. Peak voltage is not duration; duration is not calcium; calcium is not vesicle count; vesicle count is not postsynaptic current; postsynaptic current is not field power; and field power is not qualia. APME becomes meaningful only when it predicts how measured differences are transformed between these variables.

The strongest established bridge is recent-history-dependent waveform variation and its possible consequences for synaptic integration and transmission. Kv3.3 deletion provides a more recent example in which broader presynaptic spikes, increased release, altered short-term depression, and degraded auditory signal-to-noise occur in one bounded preparation [13]. Conversely, GPR55 activation can suppress release by changing vesicle responsiveness without changing the measured waveform or calcium influx [14]. This limiting case is important because APME cannot be treated as the only presynaptic transformation route. The strongest original SAN extension is the receiver-relative operator that joins this physiology to PWD, Neural Tuning, and distributed rendering. The strongest risk is overcompression: describing a complex, stochastic, compartment- specific chain as if duration directly writes a semantic quantity into a fixed vesicle alphabet. The formalism and experiments presented here are intended to replace that compression with a testable model.

Version 2 adds a further discipline: even a mathematically well-defined local sensitivity is not a biological result until the relevant variables are simultaneously measured and intervened upon. The synthetic pilot shows that a typed route can outperform rate-only, waveform-only, untyped, and larger endpoint alternatives when the nominated variables are actually present and observed. The wrong universal inverse relation then fails sharply. This supports the measurement strategy, not the biological instantiation. The decisive evidence must still come from simultaneous source, bouton, calcium, release, receiver, and population measurement with blockade and rescue.

If supported, APME would add a useful dimension to neural coding: a spike would remain an all-or-none regenerative event while its waveform and synaptic transformation carried bounded information about recent state. If unsupported, the experiments would still clarify where waveform history is lost, stabilized, or rendered redundant by simpler rate and timing codes. Either outcome improves the explanatory precision of SAN.

14 Conclusion

The Action-Potential Magnitude Effect proposes that the informational consequence of a spike cannot always be reduced to its occurrence or peak amplitude. Recent conductance history may shape the waveform; axons and boutons may transform it; calcium and release may translate it probabilistically; and receivers may interpret the result relative to their own state. SAN hypothesizes that these differences can contribute to PWDs and distributed neural rendering.

The hypothesis is deliberately narrower than the phrase *recognition changes duration*. The scientific claim is conditional, multivariate, compartment-specific, and receiver-relative. Its decisive test is whether waveform-aware mediation predicts and controls downstream neural or behavioral consequences better than established alternatives.

The version-2 contribution is a typed research program rather than a new universal magnitude law. Electrical, chemical, mechanical, and spectral observables remain distinct; their links are local sensitivities under explicit maps. The Lean kernel verifies the conditional algebra, and the application demonstrates discriminating model comparisons in synthetic systems. Neither replaces the biological experiment.

Historical Development and Authorship

I developed the core Self-Aware Networks program across work beginning in 2011. I wrote and publicly Git-fixed the APME mechanism in stages from June through August 2022, before ChatGPT's public release in November 2022. Its original theory, hypotheses, and source selections were not authored or invented by AI. Later computational tools, including AI systems, assisted under my direction with mathematical formulation, source routing, application development, formal proof implementation, grammar and spelling checks, and medical, biological, and physical accuracy review. Those tools are not independent scientific evidence and do not replace my review or external specialist review.

Data, Code, and Formal Materials

The release companion contains the frozen configuration, reference application, complete reported and deterministic-rerun tables, generated figures, manifests, validators, validation receipts, Lean source, claim signature, compatibility ledger, compiler receipt, source ledger, and claim-source matrix. The application uses synthetic data and requires no personal data.

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SAN Source Genealogy

The following sources establish SAN genealogy, not independent biological validation:

- Blumberg M. *Neural Lace Podcast 1*. Public audio, 2017; transcript Git-fixed in 2025, lines 176-229. TCP/UDP protocol question and the intentional Alternating Transmission Protocol/cellular ATP double meaning; an architectural ancestor, not the APME mechanism.
- Blumberg M. `a0089z ctpr.txt`, June 8, 2022, lines 19-33. SAN Git repository; early joined all-or-none, duration, potassium, terminal calcium, and release hypothesis.
- Blumberg M. `a0053z.md`, August 3, 2022, lines 4-6. First verified APME naming.
- Blumberg M. `a0008z.md`, August 5, 2022, lines 23-25. Sharper potassium, wave-shape, calcium-open-time, and release wording.
- Blumberg M. `a0115z.md`, July 17, 2022, lines 20-28. Population `1/f` versus single-neuron duration question.
- Blumberg M. `a0149z.md`, July 7, 2022, lines 3-15. Neuronal duration/frequency to population amplitude/frequency and phase-sequence rendering speculation.
- Blumberg M. `a0272z.md`, August 10, 2022. Umami receptor/channel/duration/release hypothesis.
- Blumberg M. `a0296zWhisker.md`, November 29, 2022, lines 5-8. Whisker state, desynchronization, coincidence, and rendered differentiation.
- Blumberg M. `a0011z.md`, January 4, 2024, lines 125-129. Mechanoreceptor/potassium/AP-duration hypothesis.
- Blumberg M. `02san.md`, September 6, 2024. PWD, tonic/phasic roles, Neural Tuning, and rendering.

Later portions of the file contain mixed dialogue and must not be treated as uniformly verbatim historical evidence.

- Blumberg M. 13san.md, June 16, 2025. Coefficient-of-variation extension; later synthesis requiring mixed-authorship caution.
- Blumberg M. Hex 7.md, January 4, 2024, line 1309. Multimodal broccoli/kale representation example; later synthesis.